

3. We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

a. the predicted value of y for $x_0 = 4$.

$$\hat{y}_i = 1.2 + 0.8x_i$$

$$\hat{y}_i = 1.2 + 0.8 * 4 = 1.2 + 3.2 = 4.4$$

b. the $se(f)$ corresponding to part (a).

$$se(f) = \sqrt{var(f)} = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]}$$

$$\hat{\sigma}^2 = \frac{SSE}{N - 2}$$

$$\hat{\sigma}^2 = 3.6 / (5 - 2) = 1.2, \bar{x} = (3 + 2 + 1 - 1 + 0) / 5 = 1, var(f) = 1.2 * [1 + 1/5 + (4 - 1)^2 / 10] = 2.52, se(f) \approx 1.5875$$

c. a 95% prediction interval for y given $x_0 = 4$.

$$\hat{y}_0 \pm t_c se(f), t_c = t_{(n-2), \frac{\alpha}{2}}$$

A 95% prediction interval for y given $x_0 = 4$ is $[-0.6520, 9.4520]$.

d. a 99% prediction interval for y given $x_0 = 4$.

A 99% prediction interval for y given $x_0 = 4$ is $[-4.8722, 13.6722]$.

e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

A 95% prediction interval for y given $x = \bar{x}$ is $[-1.8189, 5.8189]$. For Part (c), $x_0 = 4$, the interval width is $9.4520 - (-0.6520) = 10.104$. For Part (e), $x_0 = \bar{x} = 1$, the interval width is $5.8189 - (-1.8189) = 7.6378$.

The 95% prediction interval for $x_0 = \bar{x}$ is narrower than that for $x_0 = 4$. This occurs because the standard error of prediction includes the term $(x_0 - \bar{x})^2 / \sum (x_i - \bar{x})^2$. When x_0 is closer to the mean $\bar{x}$, this term is smaller, resulting in a smaller standard error and therefore a narrower prediction interval.